
Refine — Air-Gapped Setup Guide

For an architectural overview, see [NETWORK_ARCHITECTURE.md](#) (specifically Topology D).

Directory Integration in Air-Gapped Deployments

What: Refine supports LDAP / Active Directory in air-gapped deployments. Azure AD is **not supported** in air-gapped because it requires public internet egress to Microsoft endpoints.

Why:

- **LDAP / on-prem AD:** server-to-server bind. No public-internet calls. Works if your AD is reachable via VPN, Direct Connect, peering, or Transit Gateway.
- **Azure AD:** OAuth flow requires HTTPS to `login.microsoftonline.com` (or `.us` for Government), which is unreachable from a VPC with no NAT.

How:

- Configure LDAP via the in-app **Directory Settings** page after first login.
 - Verify your VPC can route to your AD server (Direct Connect / peering / Transit Gateway).
 - Verify DNS resolution for the AD hostname (Route 53 Resolver outbound endpoint to on-prem DNS, or use IP directly).
 - See [LDAP_SETUP_GUIDE.md](#) for detailed configuration steps.
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Same-Partition Limitation

A fully air-gapped Refine instance can monitor accounts **ONLY** in the same AWS partition. VPC endpoints cannot span partitions, and no public NAT egress is available.

If you need to monitor BOTH commercial AND GovCloud accounts, you have two options:

1. **Run Refine in a non-air-gapped VPC** (with NAT Gateway), use stored IAM keys for the cross-partition accounts. See [NETWORK_ARCHITECTURE.md](#) (Topology B).
 2. **Run TWO Refine instances** — one per partition, both air-gapped. No cross-instance data sharing. Out of scope for the current release.
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Required VPC Endpoints

For Topology D (fully air-gapped GovCloud), create these VPC endpoints in your VPC before deploying the CF stack. Substitute your region for `us-gov-west-1` below.

Service	Type	Endpoint name	Used by
S3	Gateway	<code>com.amazonaws.us-gov-west-1.s3</code>	Delivery ZIP download, S3 reads from monitored accounts
STS	Interface	<code>com.amazonaws.us-gov-west-1.sts</code>	Cross-account assume-role
Lambda	Interface	<code>com.amazonaws.us-gov-west-1.lambda</code>	Refine invokes the customer's collector Lambda
SSM	Interface	<code>com.amazonaws.us-gov-west-1.ssm</code>	Admin access via Session Manager
SSM Messages	Interface	<code>com.amazonaws.us-gov-west-1.ssmmessages</code>	Session Manager data plane
EC2 Messages	Interface	<code>com.amazonaws.us-gov-west-1.ec2messages</code>	Session Manager data plane
CloudWatch Logs	Interface	<code>com.amazonaws.us-gov-west-1.logs</code>	Optional
ECR API	Interface	<code>com.amazonaws.us-gov-west-1.ecr.api</code>	Only if hosting the Refine image in your private ECR
ECR DKR	Interface	<code>com.amazonaws.us-gov-west-1.ecr.dkr</code>	Same

Cost: ~\$7/mo per Interface endpoint. Gateway endpoints (S3) are free.

Custom AMI Build (Required for Topology D)

In a fully air-gapped VPC, the EC2 cannot reach `get.docker.com`, `awscli.amazonaws.com`, or any apt/dnf package repos. You must build a custom AMI with all required tools pre-installed.

Required tools in the AMI

- Docker Engine + systemd unit
- Docker Compose v2 plugin (`/usr/local/lib/docker/cli-plugins/docker-compose`)
- AWS CLI v2 (`/usr/local/bin/aws`)
- `python3`, `unzip`, `ca-certificates`, `curl`
- `amazon-ssm-agent` (already on AL2023; install via deb on Ubuntu)
- (Optional) The Refine Docker image pre-loaded via `docker load` so the EC2 can skip even the S3 download

Recommended base AMI

Amazon Linux 2023 — ships with `amazon-ssm-agent` pre-installed. Build steps:

```
# Launch a temporary EC2 from a stock AL2023 AMI in a non-air-gapped VPC
sudo dnf update -y
sudo dnf install -y docker python3 unzip awscli
sudo systemctl enable docker

# Docker Compose v2
sudo mkdir -p /usr/local/lib/docker/cli-plugins
sudo curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 \
  -o /usr/local/lib/docker/cli-plugins/docker-compose
sudo chmod +x /usr/local/lib/docker/cli-plugins/docker-compose

# Optional: pre-load the Refine image so the air-gapped EC2 skips the S3 download too
# (Only do this if you have the refine-image-*.tar from your delivery)
# sudo docker load -i refine-image-vX.Y.Z.tar

# Stop the instance and create an AMI from it
# Then copy that AMI to your air-gapped GovCloud account
```

In the CF stack, set `OperatingSystem=Custom` and `CustomAmiId=ami-xxx` (your hardened AMI). The UserData script in `aws-server-auto.yaml` v2.19 and later (current: v2.32.0) skips installs when tools are already present, so it boots cleanly with no public-internet calls.

1. What's in the Package

File	Purpose
<code>setup.bat</code> / <code>setup.ps1</code>	Windows installer
<code>setup.sh</code>	Linux / Mac installer
<code>refine.license</code>	Your signed license file — keep this private
<code>docker-compose.yml</code>	Container configuration
<code>.env.example</code>	Configuration reference
<code>refine-image-*.tar</code>	Docker image (bundled for offline use)
<code>aws-setup.yaml</code>	AWS CloudFormation template (pre-filled with your info)
<code>CHANGELOG.md</code>	What changed in this version

You will also receive an **activation code** (`RF-XXXX-XXXX-XXXX-XXXX`) through a separate channel.

What You Get

Refine analyzes **14 AWS services** across all regions and provides:

- **Rightsizing recommendations** with ready-to-run CLI commands for EC2, RDS, and more

- **Exemptions** — mark resources as intentionally sized; exempt resources are excluded from future recommendations
- **Active Savings Plans tracking** — monitor utilization, coverage, and expiration across all plans with guardrails for underutilization
- **Savings Report** — unified view of realized and projected savings across your accounts
- **Cost Cleanup** — identify orphaned EBS snapshots, unused volumes, idle NAT gateways, and other waste
- **System Events** — audit log of syncs, exemptions, and configuration changes

All data stays in your AWS account. Refine has read-only access to an S3 bucket in your environment — no data leaves your network.

2. Prerequisites

Install these **in order** before running the setup wizard.

2a. WSL 2 (Windows only — required before Docker)

Open **PowerShell as Administrator**:

```
wsl --install
```

Restart your computer after WSL finishes installing.

2b. Docker Desktop

Platform	Download
Windows	https://desktop.docker.com/win/main/amd64/Docker%20Desktop%20Installer.exe
Mac — Apple Silicon (M1/M2/M3/M4)	https://desktop.docker.com/mac/main/arm64/Docker.dmg
Mac — Intel	https://desktop.docker.com/mac/main/amd64/Docker.dmg
Linux	https://docs.docker.com/desktop/install/linux/

After installing, **open Docker Desktop** and wait for the engine to fully start before proceeding.

Air-gapped note: Download Docker Desktop on a machine with internet, copy via USB, and install offline. Docker Desktop does not require internet after installation.

2c. Python 3.8+

Platform	Download
Windows 64-bit	https://www.python.org/ftp/python/3.13.2/python-3.13.2-amd64.exe
Mac (Universal)	https://www.python.org/ftp/python/3.13.2/python-3.13.2-macos11.pkg
Linux	<code>sudo apt install python3</code> or <code>sudo dnf install python3</code>

Windows: Check **"Add Python to PATH"** on the first screen of the Python installer.

3. First-Time Installation

Step 1: Extract the ZIP

Extract to a permanent location — this is where Refine lives.

- **Windows:** `C:\Refine`
- **Linux / Mac:** `/opt/refine` or `~/refine`

Do not extract to a temporary folder or move it after setup.

Step 2: Run the Setup Wizard

Make sure Docker Desktop is running, then:

Windows — double-click `setup.bat`

Linux / Mac:

```
chmod +x setup.sh && ./setup.sh
```

The wizard will:

1. Verify Docker is running
2. Ask for your activation code
3. Prompt for an admin email and password (this becomes the ROOT account)
4. Ask for a port (default: 8000)
5. Generate a secure configuration (`.env` file)
6. Load the bundled Docker image (no internet needed)
7. Start Refine and open your browser

4. First Login

Open **`http://localhost:8000`** (or your custom port). Log in with the admin email and password you chose during setup.

You'll land on the **Setup** page to connect your AWS account. You are logged in as the **ROOT** user — the server administrator who can manage all users, groups, and AWS accounts.

Note: To add more users, go to **User Management** in the sidebar after connecting your first AWS account. See the [User Guide — User Management](#) for details.

5. Connect Your AWS Account

Step 1 — Add Your AWS Account

Enter your **12-digit AWS Account ID** and select the region for your CloudFormation stack. Click **Add Account**.

Step 2 — Deploy CloudFormation

1. Click **Download CloudFormation Template** in Refine
2. In the **AWS Console** → **CloudFormation** → **Create Stack** → **Upload a template file**
3. Upload the template, name the stack (e.g., `blacktip-refine`), and click through
4. Check the IAM acknowledgment checkbox → **Create stack**
5. Wait for **CREATE_COMPLETE** (~2-3 minutes)

Step 3 — Copy Outputs into Refine

From the CloudFormation **Outputs** tab, copy these values into Refine:

CloudFormation Output	Refine Field
RoleArn	Role ARN
CostDataBucketName	S3 Bucket Name
RefineServiceAccessKeyId	Service Access Key ID (<i>access-key mode only</i>)
RefineServiceSecretAccessKey	Service Secret Access Key (<i>access-key mode only</i>)
CredentialMode	(<i>informational — which mode the stack deployed in</i>)

The `RefineServiceSecretAccessKey` is only shown at stack creation. If you miss it, delete and recreate the stack.

Per-account credentials: Each AWS account stores its own Access Key ID and Secret Access Key in Refine's database. There are no global AWS credentials in the `.env` file. Adding or changing credentials for one account does not affect any other account.

Keyless alternative: if Refine runs on EC2 in the **same AWS partition** as the account you're onboarding, deploy the template with the `RefineHostRoleArn` parameter set to the Refine server's IAM role ARN. No IAM user, no access key, and no `RefineServiceAccessKeyId` / `RefineServiceSecretAccessKey` outputs are created — set **Access Method** to **Keyless** — this Refine server's IAM role in Refine. Keys are still required if Refine runs outside AWS or in a different partition than the monitored account.

Step 4 — Validate

Click **Validate Setup**. Refine tests IAM role assumption and S3 access. On success, an initial data sync triggers automatically.

Multiple AWS Accounts

Go to **AWS Accounts** in the sidebar → **Add Account**. Each account gets its own CloudFormation stack and its own set of service credentials — accounts are fully independent. Repeat Steps 1-4 for each additional account.

GovCloud: AWS GovCloud accounts are supported. Select the appropriate GovCloud region when adding the account.

After Setup

- **Automatic sync:** Daily at 2 AM UTC (configurable), collecting 14 AWS services in parallel across all regions
- **Manual sync:** Click **Sync Now** on any page
- **Data stays in your AWS account** — Refine has read-only access to your S3 bucket

Multi-User Access

Refine supports multiple users on a single server. All AWS accounts are loaded onto one instance, and access is controlled through user roles and account groups.

- The ROOT user (the person who ran the setup wizard) creates users from the **User Management** page and assigns each user a role and one or more account groups.
- **Account groups** are named collections of AWS accounts (e.g., "Production Team", "Dev Team"). Users see only the accounts in their assigned groups.
- Three roles: **ROOT** (sees all accounts, manages all users and groups), **ADMIN** (manages users within their assigned groups, full actions within group accounts), and **USER** (full actions within group accounts, cannot manage users).

See the [User Guide — User Management](#) for step-by-step instructions.

6. Updating

When you receive a new delivery ZIP:

1. Stop Refine (optional): `docker compose down`
2. Extract the new ZIP **into the same folder**, overwriting when prompted
3. Re-run the setup wizard — it detects your existing config and enters **update mode**:
 - Skips activation code
 - Loads the new Docker image
 - Preserves all configuration and data

Updating CloudFormation

If the changelog mentions CloudFormation changes (such as new Lambda metrics), update your existing stack — do not delete and recreate it. Updating preserves your S3 bucket and all collected data.

The new `aws-setup.yaml` is included in your updated delivery ZIP.

1. Go to **AWS Console** → **CloudFormation** → select your stack → **Update**
2. Choose **Replace current template** → upload the new `aws-setup.yaml` from the delivery ZIP
3. Click through all pages — your existing parameters (ExternalId, Email, SyncSchedule) are preserved automatically
4. Acknowledge the IAM capabilities checkbox → **Update stack**
5. Wait for status **UPDATE_COMPLETE**

Why update? The updated Lambda collects EC2 running hours across 3-month and 1-year windows. Without a stack update, those columns in the EC2 table will be empty.

Updating Your License

Replace `refine.license` and restart:

```
docker compose down && docker compose up -d
```

7. Backup & Recovery

What to Back Up

Item	Contains
<code>data/</code> folder	Database, VERSION file
<code>.env</code> file	Configuration, JWT secret
<code>refine.license</code>	Your signed license

Backup

```
cp -r data/ data-backup-$(date +%Y%m%d)/
cp .env .env.backup
cp refine.license refine.license.backup
```

Restore

```
cp -r data-backup-YYYYMMDD/ data/
cp .env.backup .env
cp refine.license.backup refine.license
docker compose down && docker compose up -d
```

Moving to a New Machine

1. Install Docker Desktop and Python on the new machine
2. Copy the entire Refine folder (including `data/`, `.env`, `refine.license`)
3. Load the image: `docker load -i refine-image-*.tar`
4. Start Refine: `docker compose up -d`

8. Troubleshooting

Problem	Solution
Refine won't start	Check Docker is running: <code>docker info</code> . Check logs: <code>docker compose logs --tail 50</code>
License expired/invalid	Contact Blacktip Solutions for a renewed <code>refine.license</code> file
Port already in use	Edit <code>PORT</code> in <code>.env</code> (e.g., <code>PORT=8080</code>), then restart
CloudFormation failed	Check the Events tab for details. Common: region not supported, missing IAM permissions
No data after sync	Check Lambda logs in AWS Console. Check S3 for <code>inventory/</code> folders. Click Sync Now in Refine

Securing Your Configuration

After first startup, restrict `.env` file permissions:

```
chmod 600 .env
```

Need Help?

Contact Blacktip Solutions at support@blacktip-ops.com